

Bharath Reddy Chada

AI/ML Engineer

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SUMMARY:

AI/ML Engineer with 8+ years of experience delivering machine learning, predictive analytics, and Generative AI solutions across enterprise technology and supply chain environments. Expertise in LLM applications, RAG architectures, NLP, real-time inference, and MLOps using Python, TensorFlow, PyTorch, LangChain, and AWS cloud technologies. Proven track record of improving model accuracy by up to 42%, accelerating deployment cycles by 60%, and driving multimillion-dollar operational savings through data-driven optimization. Combines deep technical expertise with business acumen to build scalable AI systems that enhance decision-making, efficiency, and customer outcomes.

SKILLS:

Core Strengths: Machine Learning Systems, Predictive Analytics, Generative AI Solutions, MLOps, Real-Time Inference, Data-Driven Decision Making, Supply Chain Optimization

Programming & Query Languages: Python, SQL

Machine Learning & Deep Learning: Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Time Series Forecasting

Generative AI & NLP: GPT, BERT, LLaMA, LangChain, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), Vector Embeddings, FAISS, Prompt Engineering, Fine-Tuning, Semantic Search, Conversational AI

Supply Chain Analytics & Optimization: Demand Forecasting, Inventory Optimization, Logistics Analytics, Capacity Planning

Data Engineering & Processing: Pandas, NumPy, SQLAlchemy, ETL Pipelines, Data Validation, Data Modeling, Feature Stores

MLOps & Deployment: Docker, Kubernetes, Airflow, Jenkins, CI/CD Pipelines, Model Deployment, Model Versioning, Drift Monitoring, Experiment Tracking, ML Lifecycle Management

Cloud & AI Platforms: AWS SageMaker, AWS Lambda, Amazon EC2, Amazon S3, AWS Cloud Services

Visualization & Reporting: Power BI, Tableau, Data Visualization, Dashboard Development, Business Intelligence Reporting

Tools & Collaboration: Git, GitHub, Agile, Scrum, Jira, Confluence, Cross-Functional Collaboration, Stakeholder Management

EDUCATION:

Master of Science in Business Analytics
2024

University of the Pacific

May

Bachelor of Technology in Mechanical Engineering
2017

Anurag University

Apr

EXPERIENCE:

ServiceNow, CA | May 2023 – Current | AI/ML Engineer

- Architected enterprise-scale Generative AI solutions using GPT, LLaMA, LangChain, RAG, and FAISS, improving knowledge retrieval accuracy by 42% while reducing response latency by 35% across customer support workflows.
- Developed machine learning models with XGBoost, LightGBM, TensorFlow, and PyTorch that increased prediction accuracy by 28% and enhanced business decision-making for high-volume operational use cases.
- Engineered real-time inference pipelines on AWS SageMaker, Lambda, EC2, and S3, supporting over 5M monthly predictions with 99.9% service availability and sub-second response times.
- Optimized NLP applications through prompt engineering, vector embeddings, semantic search, and fine-tuning techniques, boosting user engagement metrics by 37% and reducing manual research effort by 50%.
- Automated end-to-end ML lifecycle management using Docker, Kubernetes, Airflow, Jenkins, and CI/CD pipelines, accelerating deployment cycles by 60% while improving model reliability and governance.
- Spearheaded model monitoring and drift detection frameworks with experiment tracking and version control, reducing production model degradation incidents by 45% and strengthening operational stability.

AMAZON, India | Sep 2017 - Dec 2022 | Manager - Supply Chain & Logistics

- Led demand forecasting transformation initiatives utilizing machine learning algorithms and time-series forecasting models, improving forecast accuracy by 31%.
- Implemented predictive analytics solutions using Python, SQL, Scikit-learn, and XGBoost to identify fulfillment bottlenecks, increasing logistics efficiency by 24% across regional operations.
- Designed inventory optimization models leveraging feature engineering and statistical forecasting techniques, decreasing stockout occurrences by 27% while maintaining service-level targets above 98%.
- Established automated ETL pipelines and analytical data frameworks using Pandas, NumPy, SQLAlchemy, and AWS services, reducing reporting turnaround time by 70% for executive stakeholders.
- Piloted an ML-driven capacity planning project that analyzed transportation demand patterns and operational constraints, improving resource utilization by 22% during peak-volume periods.

- Integrated classification and anomaly detection models into supply chain monitoring processes, enabling early risk identification and reducing operational disruptions by 30%.